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# Introduction



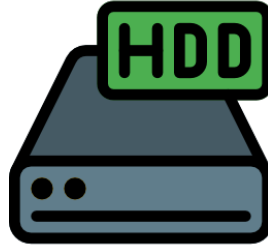
In this lesson, we are looking at the different testing and method procedures when we are assessing and inspecting computer systems and networks.

Knowing how to handle and maintain components is an essential step. Your job as a computer technician does not end with just assembling or connecting a network - it extends as well to make equipment stable for use for efficiency.

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# TOPIC 1: Testing the Hard Drive

A bad hard drive can cause more issues to your computer.



The following steps will ensure that hard drive errors can be addressed before it makes serious damages to the unit itself. Take note that the following issues can be caused by a much bigger issue other than a bad disk drive such as:

1. Errors when reading, copying, moving, or deleting data on the computer
2. Extremely slow system.
3. Operating system unable to boot.
4. Random error or computer randomly boots

## Software Solutions

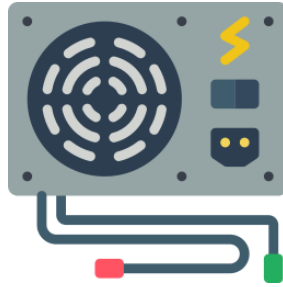
The following shows a list of software programs that can be used to test the computer's hard drive for errors:

- Scandisk
- Chkdsk
- Testdisk
- Hitachi Drive Fitness Test (DFT)
- Seagate Sea Tools
- HDD Health
- Ultimate Boot CD

## Replacing the Hard Drive

If the hard drive is bad or is generating SMART error (A S.M.A.R.T. error is a predictive indicator that there will be a drive failure.) If you get a S.M.A.R.T. error, it means that you have to back-up since you would need to replace the hard drive soon. If the computer is relatively new, the hard drive may still come under a warranty.

## TOPIC 2: Testing the Power Supply Unit



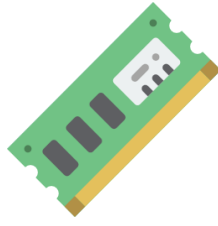
Some programs such as SpeedFan can help monitor the voltage and power supply to computer fans; however, there is no software utility to test the integrity of computer power supplies.

**It is recommended to replace the Computer Power Supply if it is no longer usable.**

It is also wise to check how much wattage your computer really requires. You can do this online at a PSU Wattage Calculator. This is to provide a more efficient electricity usage when checking your PSU.

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# TOPIC 3: Testing the Memory



The following are some indicators that your computer's memory may already be bad:

1. The computer does not boot and a beeping sound is heard.
2. There are random computer crashes causing the Blue Screen of Death (BSOD), General Protection Fault Error Messages, Illegal Operations, Fatal Exceptions, etc.
3. Computer randomly reboots
4. Windows or other program installation failure.

*Take note that before testing your memory, make sure that you have removed recently added new memory to make sure that it is not the cause of the issue.*

## Software Solutions

Several software programs are available to test computer hardware, including computer memory. These include:

### • Windows Memory Diagnostic Tools

It can be accessed on Windows Vista and Windows 7. Simply search for "Memory" on the Start Search Bar. Doing so will open the Memory Diagnostics Tool. Two options will appear: (1) Restart Now and Check for Problems and (2) Check for problems next time I start my computer. Simply choose one of these options to run a memory test on your preferred time.

### • Memtest86

An x86 compatible, free memory tester

### • DocMemory

A software that can be used to check the computer's memory

There are other software programs that can be used to check your computer's memory.

Some are used at computer service centers or may be issued by the company you are working for.

Other software programs that can be used are:

- PC-doctor
- PC Diagnostics
- Ultra X

## Swap and Remove Memory

If there is an available compatible memory from another computer, you can try swapping the memory into the computer to check whether it is the issue.

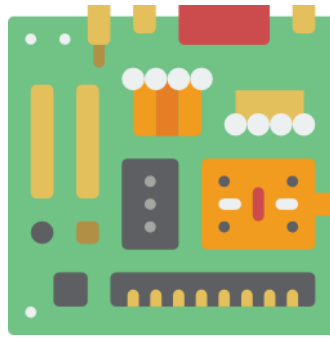
If the memory works in another computer without any issues, there may be a problem with the motherboard, PSU, or other hardware issue.

## Replace Memory

If computer memory is completely bad already, replace it with a new one.

# TOPIC 4: Testing the Motherboard and CPU

A bad computer motherboard or CPU can cause different issues on a computer.



Below are some of the issues that you may encounter due to a computer with bad motherboard and CPU:

1. The computer does not boot, but instead gives a beep code.
2. The computer randomly crashes causing General Protection Fault error messages, illegal operations, Fatal exceptions, etc.
3. The computer suddenly reboots.

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Here are some recommendations to test the computer's motherboard and CPU:

## Visual Hardware Check

Check the computer motherboard if there is a presence of bulged or blown capacitors.

Check if there is a slight bulge on the top of each capacitor. If the capacitor is leaking, it can be an indicator that it has already blown. Blown capacitors are likely caused by motherboard issues in the computer.

Once the CPU is removed, check for bent pins on the side that are inserted into the motherboard. If there is a bent pin, it is most likely the cause of the computer's CPU issues.

It is possible to replace a blown or bulging capacitor, but it will require precision soldering to install the capacitor.

Bent pins on a CPU can be bent back into place, but make sure to be very careful because the pin might break off when trying to bend it back into place. If the pin breaks off, the CPU will definitely need to be replaced.

If you are done with the visual hardware check and there seems to be no obvious issues, there may be an issue with the software.

## Software and Hardware Solutions

Here are some software programs that you can use to test your computer's motherboard and CPU. However, take note that problems related to the motherboard and CPU have a wide variety of complexity; thus, these programs may not detect every possible failure.

- **Hot CPU Tester** – This tester includes a burn-in feature for new computers or computers with new motherboard or CPU.

- **Intel Processor Diagnostic Tool** – This can be used for computers running on Intel processor. It is a free utility for testing a variety of Intel processors.

- **PC-doctor** – A good, but quite pricey software used by service centers and technicians to diagnose computer hardware issues including motherboard issues.

- **Ultra-X** – A collection of products that can be used to test computer hardware, including the motherboard.

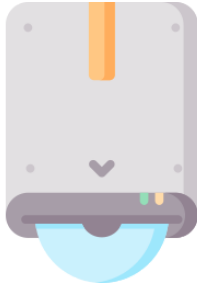
## Replacing the Motherboard and CPU

After you are done with the visual hardware check and software/hardware check and it was determined that one or both can no longer be used, then it is already time to replace the motherboard and the CPU. There is typically no easier or cheaper alternative to fixing the motherboard or the CPU.

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# TOPIC 5: Testing the Optical Drive



Before testing this, make sure that multiple discs do not work in the drive. If only one CD or DVD is causing the problem, consider that it can be an issue with the CD and not the CD/DVD drives.

A bad disc drive can cause a variety of issues to your computer. Here are some issues that are caused by a bad drive:

1. Errors when reading CD or DVD.
2. CDs or DVDs may not play the audio or video properly.
3. CD or DVD programs may not install or may encounter errors after being installed.

Here are some recommendations to test your computer's CD ROM and DVD drive:

## Software and Hardware Solutions

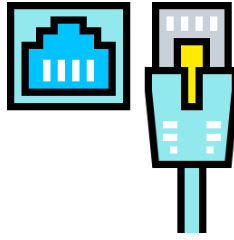
The most common issue that CD ROMs and DVD drives is caused by dust and dirt making it unable for the drives to read discs. You can use a CD/DVD Drive Cleaner kit to help clean the disc drive. These can be purchased in many office supply stores and electronic stores. Some software tools that can be used are:

- **CD Roller** – Program that can be used to test CD/DVD drives as well as recover data from CDs and DVDs
- **CD Check** – Another program that can be used to check CD drives (no DVD compatibility) and can also recover data from damaged disks.
- **CD/DVD Diagnostic** - Diagnostic programs from InfinaDyne that is not only capable of testing a drive and disc, as well as recover data from disks that have gone bad.

## Replace the Disc Drive

After trying various software and hardware solutions and the disc drives still appear to be bad, it is highly recommended to replace the disc drives.

# TOPIC 6: Testing the Network



**After setting up network switches, cables, and configuration, verify whether the computer network is functional.**

*Here are some steps that you can do to check your computer connection:*

## **Check physical connections**

Check whether the little red or green light next to the RJ-45 port is lit on every computer. Check this light both on the computer itself and on the switch or router the computer is plugged into.

If the light is not on, you may have a connection problem—most likely a poorly connected cable.

## **Verify that you can log on**

Try logging on to each of your network computers using a valid domain user account.

## **Check the network configuration**

Launch the Command Prompt then enter the command `ipconfig / all` and press Enter. This command will give out lines of information. In order to make sure that your computer network is working properly, make sure that this line is present:

IPv4 Address. . . . .: 192.168.1.125 (Preferred)

If this line does not show, check that your IP Configuration is set correctly and the DHCP server is working.

## **Verify that computers can ping each other**

You can try using the ping command from the command prompt to make sure that all computers on the network can contact one another. Make sure that the TCP/IP is up and running by having the computer ping itself. Launch the Command Prompt and type `ping localhost`. The prompt will indicate whether the ping was successful.